

The Foundations: Logic and Proofs

Chapter 2, Part III: Proofs

Rules of Inference

Section Summary

- Valid Arguments
- Inference Rules for Propositional Logic
- Using Rules of Inference to Build Arguments
- Rules of Inference for Quantified Statements
- Building Arguments for Quantified Statements

What is an Argument in Logic?

- An **argument** is a sequence of statements where:
 - The **statements before the last one** are called **premises** (facts or assumptions).
 - The last statement is the **conclusion** (what we want to prove).
- **Example:**
 - It is raining.
 - If it is raining, the road is wet.
 - Therefore, the road is wet.
- **First two** → **premises**
- **Last one** → **conclusion**

Valid Argument

An argument is **valid** if:

Whenever all the **premises** are **true**, the **conclusion** must **also be true**.

So:

- Premises **TRUE**
- Conclusion **FALSE**
- This situation must be **IMPOSSIBLE**

Example-VALID argument

If I study, I pass.

I study.

Therefore, I pass.

This **cannot fail** → **VALID**

INVALID argument

If I study, I pass.

I pass.

Therefore, I studied.

Could pass by luck →

INVALID

Argument – Example

- Consider the following argument (sequence of propositions)
- *“If you have a current password, then you can log onto the network.”*
- *“You have a current password.”*
- *Therefore, “You can log onto the network.”*
- To determine if it is a valid argument?

Argument Form

- An **argument form** is a general pattern of an argument using variables $(p, q, r, \dots)$.
- An argument form is **valid** if it remains valid **no matter what statements replace the variables**.

Logical Representation

- If the premises are: $p_1, p_2, \dots, p_n$ and the
- conclusion is: q
- Then the argument is valid **if and only if**:
- $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is a **tautology** (always true).

Inference Rules

- Safe reasoning steps

that tell us **what we are allowed to conclude** from what we already know

- Example:

- If I press the switch, the light turns on.
- I press the switch.
- **Conclusion:** So, The light turns on.

- Always correct

- Never fails.

- Inference rules tell us **what we are allowed to conclude, without guessing.**

Rules of Inference

- These rules of inference can be used as **building blocks** to construct more complicated valid argument forms

Rules of Inference

TABLE 1 Rules of Inference.

<i>Rule of Inference</i>	<i>Tautology</i>	<i>Name</i>
$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	$p \rightarrow (p \vee q)$	Addition
$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	$(p \wedge q) \rightarrow p$	Simplification
$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

- We will show how to construct valid arguments in two stages; first for **propositional logic** and then for **predicate logic**. The rules of inference are the essential building block in the construction of valid arguments.

1. **Propositional Logic**

Inference Rules

2. **Predicate Logic**

Inference rules for propositional logic plus additional inference rules to handle variables and quantifiers.

Rules of Inference for Propositional Logic: Modus Ponens

- **Modus Ponens** is a basic rule of logical reasoning.
It tells us **how to reach a conclusion from two true statements.**

- **Structure:**

- $p \rightarrow q$ (If p is true, then q is true)

- p (p is true)

Conclusion: q is true

Rules of Inference for Propositional Logic: **Modus Ponens**

Symbol $\therefore$ denotes “therefore.”

$$\frac{p \rightarrow q \quad p}{\therefore q}$$

Corresponding Tautology:

$$(p \wedge (p \rightarrow q)) \rightarrow q$$

Example:

Let p be “It is snowing.”

Let q be “I will study discrete math.”

“If it is snowing, then I will study discrete math.”

“It is snowing.”

“Therefore, I will study discrete math.”

Example:

- If I study, I'll pass.
- I studied.
- Therefore, I'll pass.

•Example:

- If it rains, I take an umbrella
It is raining
Therefore, I take an umbrella

$$p \rightarrow q$$

$$\frac{p}{\therefore q}$$

If p happens, then q happens
p happens
So q happens

Modus Tollens

If p happened, q would have happened
q did NOT happen
So p did NOT happen

$$p \rightarrow q$$

$$\neg q$$

$$\therefore \neg p$$

Corresponding Tautology:

$$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$$

Example:

Let p be “it is snowing.”

Let q be “I will study discrete math.”

“If it is snowing, then I will study discrete math.”

“I will not study discrete math.”

“Therefore, it is not snowing.”

Example:

If I studied, I would pass

I did NOT pass

Therefore, I did NOT study

$$p \rightarrow q$$

$$\neg q$$

$$\therefore \neg p$$

Example:

- If it's a cat, it says "meow."
- It does not say "meow."
- Therefore, it's not a cat.

Hypothetical Syllogism

p leads to q
q leads to r
So p leads to r

$$p \rightarrow q$$

$$q \rightarrow r$$

$$\therefore p \rightarrow r$$

Corresponding Tautology:

$$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

Example:

Let p be “it snows.”

Let q be “I will study discrete math.”

Let r be “I will get an A.”

“If it snows, then I will study discrete math.”

“If I study discrete math, I will get an A.”

“Therefore, If it snows, I will get an A.”

Example:

If I exercise, I become fit

If I become fit, I stay healthy

Therefore, if I exercise, I stay healthy

$$p \rightarrow q$$

$$q \rightarrow r$$

$$\therefore p \rightarrow r$$

Disjunctive Syllogism

$$p \vee q$$

Either p or q is true
p is false
So q must be true

$$\frac{\neg p}{\text{---}}$$

Corresponding Tautology:

$$(\neg p \wedge (p \vee q)) \rightarrow q$$

$$\therefore q$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study English literature.”

“I will study discrete math or I will study English literature.”

“I will not study discrete math.”

“Therefore , I will study English literature.”

Example:

I will drink tea or coffee

I will not drink tea

Therefore, I will drink coffee

$$p \vee q$$

$$\neg p$$

$$\therefore q$$

Example:

- It is either Monday or Tuesday.
- It is not Monday.
- Therefore, it is Tuesday.

Addition

If p is true, then “ p or anything” is true

$$\frac{p}{\therefore p \vee q}$$

Corresponding Tautology:

$$p \rightarrow (p \vee q)$$

Example:

Let p be “I will study discrete math.”

Let q be “I will visit Las Vegas.”

“I will study discrete math.”

“Therefore, I will study discrete math or I will visit Las Vegas.”

Simplification

p and *q* are true
Then *p* is definitely true

$$\frac{p \wedge q}{\therefore p}$$

Corresponding Tautology:
 $(p \wedge q) \rightarrow p$

Example:

Let *p* be “I will study discrete math.”

Let *q* be “I will study English literature.”

“I will study discrete math and English literature”

“Therefore, I will study discrete math.”

Example:

- I have a pen and a book.
- Therefore, I have a pen.

Example:

I studied math and physics
Therefore, I studied math

$$\frac{p \wedge q}{\therefore p}$$

Conjunction

p

q

$\therefore p \wedge q$

Corresponding Tautology:

$$((p) \wedge (q)) \rightarrow (p \wedge q)$$

p is true
 q is true
So both are true

Example:

Let p be “I will study discrete math.”

Let q be “I will study English literature.”

“I will study discrete math.”

“I will study English literature.”

“Therefore, I will study discrete math and I will study English literature.”

Example:

I passed math

I passed physics

Therefore, I passed math and physics

Example:

- It is sunny.
- It is warm.
- Therefore, it is sunny and warm.

$$\frac{p}{q}$$

$$\therefore p \wedge q$$

Resolution

$$\neg p \vee r$$

$$p \vee q$$

$$\therefore q \vee r$$

Either not p or q

Either p or r

p is cancelled

One of q or r must be true

Corresponding Tautology:

$$((\neg p \vee r) \wedge (p \vee q)) \rightarrow (q \vee r)$$

Example:

Let p be “I will study discrete math.”

Let r be “I will study English literature.”

Let q be “I will study databases.”

“I will not study discrete math or I will study English literature.”

“I will study discrete math or I will study databases.”

“Therefore, I will study databases or I will study English literature.”

Example:

- It is not raining or I have an umbrella.
- It is raining or I am indoors.
- Therefore, I have an umbrella or I am indoors.

$$\frac{\neg p \vee r \quad p \vee q}{\therefore q \vee r}$$

Using the Rules of Inference to Build Valid Arguments

- A *valid argument* is a sequence of statements.
- Each statement is a premise.
- The last statement is called conclusion.
- A **valid argument** is built like this:
 1. Write premises
 2. Apply rules step by step
 3. Reach conclusion
- A valid argument takes the following form:

$$S_1 \quad S_2 \quad \dots \quad S_n \quad \therefore \quad C$$

Valid Arguments

Example 1: From the single proposition

$$p \wedge (p \rightarrow q)$$

Show that ***q*** is a conclusion.

Solution:

Step	Reason
1. $p \wedge (p \rightarrow q)$	Premise
2. p	Simplification using (1)
3. $p \rightarrow q$	Simplification using (1)
4. q	Modus Ponens using (2) and (3)

Example 2:

- With these hypotheses:
 - “It is not sunny this afternoon and it is colder than yesterday.”
 - “We will go swimming only if it is sunny.”
 - “If we do not go swimming, then we will take a canoe trip.”
 - “If we take a canoe trip, then we will be home by sunset.”
- Using the inference rules, construct a valid argument for the conclusion:
 - “We will be home by sunset.”

Solution:

1. Choose propositional variables:
 - p : “It is sunny this afternoon.” r : “We will go swimming.”
 - t : “We will be home by sunset.”
 - q : “It is colder than yesterday.” s : “We will take a canoe trip.”
2. Translation into propositional logic:

Hypotheses: $\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s, s \rightarrow t$

Conclusion: t

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This argument form shows that the premises lead to the desired conclusion.

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

Valid Arguments

Example 3:

● With these hypotheses:

“If you send me an e-mail message, then I will finish writing the program,” “If you do not send me an e-mail message, then I will go to sleep early,” and “If I go to sleep early, then I will wake up feeling refreshed”

lead to the conclusion:

“If I do not finish writing the program, then I will wake up feeling refreshed.”

Solution:

1. Choose propositional variables:

p: “You send me an e-mail message,” *q*: “I will finish writing the program,”

r: “I will go to sleep early,” *s*: “I will wake up feeling refreshed.”

Hypothesis: $p \rightarrow q, \neg p \rightarrow r, r \rightarrow s.$

Conclusion: $\neg q \rightarrow s$

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Hypothesis: $p \rightarrow q, \neg p \rightarrow r, r \rightarrow s.$

Statement 1:

If you send me an e-mail, I will finish the program.

So clearly:

If the program is NOT finished, then no e-mail was sent.

Statement 2:

If you do not send me an e-mail, I go to sleep early.

So now we have:

Program not finished $\rightarrow$ no e-mail $\rightarrow$ sleep early

Statement 3:

If I go to sleep early, I wake up feeling refreshed.

So now we extend the chain:

Program not finished $\rightarrow$ no e-mail $\rightarrow$ sleep early $\rightarrow$ wake up refreshed

Final conclusion

If I do not finish writing the program, I will wake up feeling refreshed.

$\neg q \rightarrow \neg p \rightarrow r \rightarrow s$

Valid Arguments

● Given hypotheses:

1. “It is raining today and I have an exam tomorrow.”
2. “I will watch a movie only if I do not have an exam tomorrow.”
3. “If I do not watch a movie, then I will study.”
4. “If I study, then I will pass the exam.”

lead to the conclusion:

“I will pass the exam.”

Solution:

- **p** : “It is raining today”
- **q** : “I have an exam tomorrow”
- **r** : “I will watch a movie”
- **s** : “I will study”
- **t** : “I will pass the exam”

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Step 2: Hypothesis

1. *It is raining today and I have an exam tomorrow*

$$p \wedge q$$

2. *I will watch a movie only if I do not have an exam tomorrow*

$$r \rightarrow \neg q$$

3. *If I do not watch a movie, then I will study*

$$\neg r \rightarrow s$$

4. *If I study, then I will pass the exam*

$$s \rightarrow t$$

Step 3: Reasoning

From $p \wedge q$

→ we know q is true (I have an exam)

• From $r \rightarrow \neg q$ and q is true

→ I **cannot** watch a movie → $\neg r$

• From $\neg r \rightarrow s$

→ I will study → s

• From $s \rightarrow t$

→ I will pass the exam → t

Final conclusion: t
“I will pass the exam.”